Detecting Sensor Attacks on Autonomous Vehicles through Latent Correlations

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Abstract: Autonomous vehicles (AVs) rely on sensors for state perception, and an attacker who corrupts even one of them can drive the vehicle off course. Most existing detectors catch such attacks by predicting how the vehicle should behave and alarming when observations disagree, so their accuracy is bounded by the accuracy of the prediction model and degrades under model mismatch, hard maneuvers, or a stealthy adversary who keeps the residual small. We present LCM, a **L**atent **C**orrelation-based **M**onitor which watches the physical coupling among the sensors themselves without relying on any prediction model. LCM projects each sensor stream onto its Karhunen–Loève basis, tracks the Pearson correlation between the leading coefficients with an EWMA control chart. Across 900 CARLA missions covering bias, replay, delay, and denial-of-service (DoS) attacks, LCM detects attacks with a mean delay of 0.498 s without any false positives, and achieves an AUC of 0.949; under an identical protocol it clearly outperforms SAVIOR, a state-of-the-art physics-based detector, reaching an AUC of 0.949 against SAVIOR's 0.75 per channel; SAVIOR's only higher-AUC configuration raises false alarms on over half of the benign segments. On a real rover under GPS spoofing it alarms within seconds of onset. Even against a white-box adversary that shapes its injection to evade detection, LCM still catches 94 of 100 attacks; on the few that evade, the deviation reaches a median of 7.449 m.

Keywords: Autonomous vehicle security, Sensor attack detection, Karhunen–Loève expansion, Correlation monitoring

1. Introduction

Autonomous vehicles (AVs) are a major step forward for intelligent transportation [1], [2], [3]. By removing the human error behind tens of thousands of road deaths each year, they could save the U.S. economy up to 190 billion dollars annually [4], and their uses extend beyond transportation to agriculture, healthcare, and space exploration [5]. At the same time, AVs rely heavily on complex technology, and failures do happen: as of January 10th, 2025, the California Department of Motor Vehicles (DMV) has received 778 autonomous vehicle collision reports [6], with sensor faults identified as a notable cause of missed obstacles, miscalculated distances, and undetected traffic elements [7].

Among the various security threats faced by AVs, sensor attacks, such as spoofing and transduction, stand out because they target the perception layer that every downstream decision depends on. Spoofing attacks can trick cameras with fake traffic signals or decoy pedestrians [8], redirect AV navigation via GPS spoofing [9], or exploit LiDAR occlusion vulnerabilities with an 80% success rate [10]. In transduction attacks, out-of-band signals such as acoustic and electromagnetic injections can compromise inertial sensors [11], [12], [13], [14]. Therefore, detecting such attacks is the first step toward protecting AVs. In this work, we consider a threat model where an adversary attacks the sensors of a victim AV to manipulate its behavior.

A variety of detection methods have been proposed for sensor attacks on AVs. CI-based detector [15] predicts vehicle behavior with a state-space model and monitors deviations over a sliding window. SAVIOR [16] builds its predictions on an extended Kalman filter (EKF), and AVMon [17] extends this design by adding a learned residual-estimation model. PID-Piper [18] approximates the controller's outputs with a machine learning model and applies CUSUM to separate attack-induced divergence from natural drift. They all raise alarms from residuals against a model's prediction of vehicle behavior. The detection quality is therefore tied to the prediction quality: when model mismatch or parameter drift, say from tire wear or payload changes, corrupts the predictions, the residuals lose their meaning, and alarms fire late, fire falsely. In addition, traditional countermeasures such as software security, authentication, and cryptography are not sufficient to defend against transduction attacks, which exploit unintended responses of sensor circuitry to compromise the integrity and availability of sensor output [16], [19].

To fill this gap, we propose a detector called **LCM**, a Latent Correlation-based Monitor for sensor attack detection on AVs. LCM performs correlation analysis in a low-

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rank, noise-robust *Karhunen–Loève Expansion* (KLE) domain [20], [21], [22] to track spatial relationships among sensors, quantified using the *Pearson Correlation Coefficient* (PCC) [23], [24], and uses an *Exponentially Weighted Moving Average* (EWMA) control chart [25], [26], [27], [28] to monitor their temporal evolution. This correlation-based detector does not rely on accurate prediction models.

We evaluate LCM both in a high-fidelity simulator and on a real-world dataset. In the CARLA simulator [29], we run 900 AV missions covering bias, replay, delay, and DoS attacks on the GNSS, wheel encoder, gyroscope, accelerometer, and compass. To check that the method works outside simulation, we also test it on the dataset of Abrar et al. [30], recorded during real GPS spoofing attacks on a physical rover. We further construct an adaptive stealthy attack tailored to evade it, and find that the vehicle's trajectory deviation stays under 2.1 meters, so physical safety holds even against this adversary. Our scope is the inertial and positioning sensors listed above and attacks on the actuator bus or the perception layer call for different defenses and are outside this paper.

2. Background and Motivation

2.1 KLE and Low-dimensional Latent Space Projection

Running correlation analysis directly on raw sensor signals has two problems: the signals are noisy, and the computation is expensive. What we want is a smaller representation that keeps the dominant structure of each signal. LCM obtains one by projecting the raw data onto a *Karhunen–Loève Expansion* (KLE) basis [20], [21], [22]. The KLE decomposes each sensor stream into orthogonal components ordered by their energy, so keeping only the leading components captures the low-rank structure of the signal in a handful of coefficients. Correlation analysis on these coefficients is cheap enough for real-time use.

This choice rests on two observations from prior work. First, multi-modal driving data carries strong spatial and temporal redundancy and is therefore well approximated by low-rank models [31], [32]. Second, the KLE has a long record in signal denoising and compression [33], [34], [35]: projecting onto the leading basis keeps the signal and sheds much of the noise. Both properties are exactly what correlation monitoring needs, so we use the KLE projection as the front end of our detection pipeline. As far as we know, this is the first use of KLE-based latent modeling for security monitoring on AVs.

2.2 PCC-Based Correlation Monitoring

Once the signals live in the KLE latent space, we need a way to measure how strongly two sensors move together, and to follow that relationship over time. LCM uses the *Pearson Correlation Coefficient* (PCC) between the KLE projections of each sensor pair. The PCC is a statistical metric that quantifies the degree of linear dependence between two signals. Formally, for signals $x = \{x_1, \ldots, x_n\}$ and $y = \{y_1, \ldots, y_n\}$, the PCC is defined as:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\,\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}, \tag{1}$$

where $\bar{x}$ and $\bar{y}$ denote the mean values of the respective signals. By analyzing correlation patterns in the transformed KLE space, LCM robustly identifies deviations indicative of sensor-level faults or adversarial perturbations. The value of r lies in the interval $[-1, 1]$, where: $r = 1$ indicates a perfect positive linear correlation, $r = -1$ indicates a perfect negative linear correlation and $r = 0$ implies no linear correlation. In the context of sensor consistency analysis for AVs, deviations in PCC values from expected correlation serve as indicators of sensor faults, attacks, or adversarial perturbations, providing a principled and efficient mechanism to detect abnormal inter-sensor behavior in real time.

PCC itself is not new to vehicular anomaly detection, but the existing detectors break down when conditions change. Sharma et al. [36] compare cross-vehicle PCC matrices and flag a vehicle as misbehaving when a threshold score reaches $h \geq 1$. However, the score is fragile against timing misalignment and sensor latency, and false positives climb as a result. Guo et al. [37] compute PCC over local windows and add a frequency-domain check on power spectral density, alarming when the energy in a fixed band crosses a fixed threshold. However, a static threshold cannot follow the non-stationarity that comes with sensor noise, maneuvers, and changing environments. LCM replaces the static threshold with an *Exponentially Weighted Moving Average* (EWMA) control chart [25], [26], [27], [28], which tracks each pair's correlation against an adaptive baseline instead of a constant.

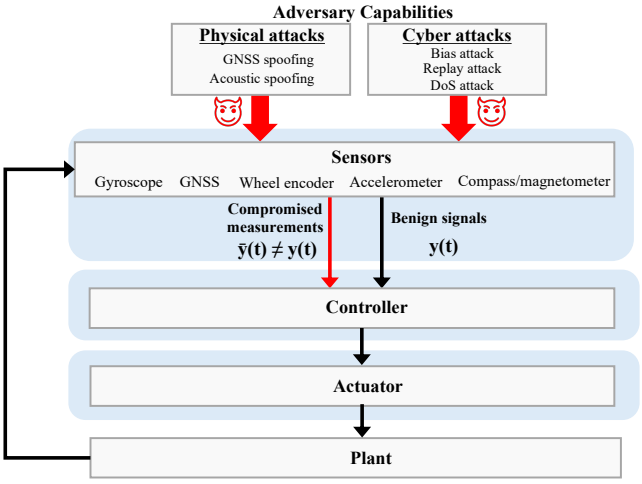


Fig. 1 Threat Model

3. Threat Model

3.1 Threat Overview

Figure 1 illustrates our threat model. The adversary tampers with onboard sensor measurements so that the vehicle misjudges its own physical state: one or more sensors report

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$\bar{y}(t)$ instead of the true value $y(t)$.

3.2 Adversarial Capabilities

We assume a white-box adversary who knows the vehicle's models, controllers, and sensor interfaces, and can use real-time information to shape the attack [38], [39]. This is the standard setting in prior work [38], [39], [40], [41], [42], and it is also the harshest one for a defender: a white-box attacker can craft stealthy attacks, so whatever detection performance survives this setting is a worst-case bound rather than an optimistic estimate.

The adversary can compromise one or several sensors at once. We focus on the inertial and positioning sensors, namely the gyroscope, wheel encoder, accelerometer, compass, and GNSS receiver, because these feed directly into state estimation and control [43]. Software- and firmware-level attacks are out of scope, since established mechanisms already address them [44], [45]. Perception sensors such as cameras, Radar, and LiDAR are assumed to work correctly. Attacks on them are a different problem with dedicated defenses [46], [47], [48], and processing their data (SLAM, deep learning) demands computing power that industrial AVs, agricultural robots, and small delivery drones often do not have [49]. LCM therefore stays on the localization and control layer, monitoring the IMU, GNSS, and wheel encoder that constrained platforms navigate by [50], without assuming full-stack sensor fusion.

3.3 Attack Types

We model four attack classes that cover the common ways a sensor reading can be falsified [51], [52], [53].

- **Bias attacks.** An offset is added to the reading: $y_t^{\text{attack}} = y_t + b$, where $b \in R$ may be constant or grow and shrink linearly over time. The measurement stays plausible while drifting away from the truth.
- **Replay attacks.** The live reading is swapped for an old one: $y_t^{\text{attack}} = y_{t'}$ with $t' < t$. The controller sees values that once were genuine, and the current dynamics are hidden behind them.
- **Delay attacks.** Readings arrive late. The adversary buffers the data and releases it after a fixed lag Δt, so $y_t^{\text{attack}} = y_{t-\Delta t}$, and the controller reacts to a state the vehicle has already left.
- **Denial of Service (DoS) attacks.** Readings stop arriving at all. The adversary blocks transmission over a set of time steps $\mathcal{T}_{\text{DoS}}$, and the controller receives zeros: $y_t^{\text{attack}} = 0$ for $t \in \mathcal{T}_{\text{DoS}}$.

4. LCM Design and Overview

In this section, we describe the architecture of LCM. We detect cross-sensor attacks by watching whether two sensor streams, $x_t^{(1)}$ and $x_t^{(2)}$, stay decorrelated after each is projected onto its KLE mode. The basis for that projection is fitted once, on benign data, as follows. Given an embedding dimension L, each stream is unrolled into a delay-coordinate matrix

$$X_{\text{train}}^{(j)} = \begin{bmatrix} x_0^{(j)} & x_1^{(j)} & \cdots & x_{L-1}^{(j)} \\ x_1^{(j)} & x_2^{(j)} & \cdots & x_L^{(j)} \\ \vdots & \vdots & \ddots & \vdots \end{bmatrix}, \tag{2}$$

whose rows are the training windows. With $\bar{\mathbf{x}}^{(j)} = \frac{1}{N_{\text{tr}}} \sum_t X_{\text{train}}^{(j)}(t,:)$ the mean over the N_{tr} windows, we center the matrix, $\tilde{X}_{\text{train}}^{(j)} = X_{\text{train}}^{(j)} - \mathbf{1}\,\bar{\mathbf{x}}^{(j)\top}$, and take the sample covariance

$$R^{(j)} = \frac{1}{N_{\text{tr}} - 1} \tilde{X}_{\text{train}}^{(j)\top} \tilde{X}_{\text{train}}^{(j)}. \tag{3}$$

Its eigen-decomposition $R^{(j)} = V^{(j)} \Lambda^{(j)} V^{(j)\top}$ gives the KLE basis $\{\phi_i^{(j)}\}$ as the columns of $V^{(j)}$. The basis is fitted once and never updated.

Quadrature-Based Projection. At runtime, every incoming window of length L is centered by $\bar{\mathbf{x}}^{(j)}$ and projected onto the basis. The inner product is approximated by Legendre–Gauss quadrature: with nodes and weights $\{\xi_q, w_q\}_{q=1}^{Q}$ mapped onto $\{0, \ldots, L-1\}$, the coefficient of mode i is

$$c_i^{(j)}(t) \;\approx\; \sum_{q=1}^{Q} w_q\, \tilde{x}_t^{(j)}(\xi_q)\, \phi_i^{(j)}(\xi_q), \tag{4}$$

where $\tilde{x}_t^{(j)}$ is the interpolated window. Of all the coefficients, we keep only the leading one, $c_1^{(j)}$, because that is where the attack signature lives: the leading mode carries the inter-sensor coupling, which comes from the vehicle kinematics.

Correlation Estimation. Over an analysis window of length W, sliding in steps of S, we compute the PCC between the leading coefficients of the two streams,

$$r_k = \operatorname{corr}\Big(\{c_1^{(1)}(t)\}_{t=k}^{k+W-1},\; \{c_1^{(2)}(t)\}_{t=k}^{k+W-1} \Big). \tag{5}$$

Raw correlations are awkward to monitor directly: their variance depends on the correlation level itself, and their sign is irrelevant for our purpose. The Fisher transform fixes the former and the absolute value the latter,

$$z_k = \big|\operatorname{atanh}\big(\operatorname{clip}(r_k, -1{+}\epsilon,\, 1{-}\epsilon)\big)\big|. \tag{6}$$

From the training windows we estimate a location and scale that outliers cannot drag around,

$$\mu = \operatorname{median}\{z_k^{\text{train}}\}, \qquad \sigma = 1.4826 \cdot \operatorname{median}\{|z_k^{\text{train}} - \mu|\}. \tag{7}$$

EWMA Smoothing and Normalization. A single decorrelated window may be noise; a run of them is an attack. To reward persistence, the sequence z_k is smoothed by an EWMA and normalized by its steady-state deviation,

$$\begin{aligned} S_k &= \lambda z_k + (1-\lambda) S_{k-1}, \quad S_0 = \mu, \\ q_k &= \frac{|S_k - \mu|}{\sigma_{\text{ewma}}}, \qquad \sigma_{\text{ewma}} = \sqrt{\frac{\lambda}{2-\lambda}}\,\sigma. \end{aligned} \tag{8}$$

An alarm is raised when $q_k \geq \kappa$, with κ set to the 90th percentile of $\{q_k^{\text{train}}\}$ on the training set.

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Table 1 Attack scenarios for LCM's evaluation, spanning a progression from single-sensor (S1) to multi-sensor, multi-type compromises (S2–S3). Each scenario is instantiated as 300 randomized CARLA missions with stochastic onset, trajectory shape, and initial speed.

Scen.	Timing	Attack Type(s)	Target Sensor(s)	Manipulated Parameters
S1	Early	Fixed bias injection	GNSS	Latitude, Longitude
S2	Middle	Increasing bias + DoS	GNSS, Compass, Wheel Encoder	Latitude, Longitude, Speed, Orientation
S3	Late	Replay + Delay + DoS	Accelerometer, Gyroscope, Wheel Encoder	Acceleration, Yaw rate, Speed

5. Evaluation

5.1 Experimental Setup

We first evaluate LCM in simulation, across 900 missions and 3 different types of attack scenarios and summarized in Table 1. The attacks target ten states and sensor channels, namely latitude, longitude, speed, orientation, accelerometer.x/y/z, and gyroscope.x/y/z, spanning the five sensing modalities (GNSS, wheel encoder, compass, accelerometer, and gyroscope). Each mission is attacked at an early, middle, or late injection time, with the four attack classes: bias (fixed and varying), replay, delay, and DoS. Time is discretized at a sampling period of $\Delta T = 0.05\,\mathrm{s}$.

We use two metrics: Attack Detection Delay (ADD) for timeliness and Receiver Operating Characteristic (ROC) for accuracy. ADD is the time between attack onset and the first alarm. With p_{attack} and p_{detect} the time steps at which the attack starts and the detector fires, the average over all missions is

$$\overline{\mathbf{ADD}} = \frac{1}{N_{\mathrm{missions}}} \sum_{j=1}^{N_{\mathrm{missions}}} \left(p_{\mathrm{detect}}^{(j)} - p_{\mathrm{attack}}^{(j)}\right) \Delta T. \quad (9)$$

The smaller the $\overline{\mathrm{ADD}}$, the less time an attack has to push the vehicle off course before anyone notices. The ROC is computed from the anomaly scores produced by LCM. Each analysis window is classified as attacked or benign by comparing its score against a threshold; sweeping the threshold and comparing the predictions with the ground-truth labels yields the true positive rate (TPR) and false positive rate (FPR) at every operating point. The area under the curve (AUC) summarizes the result, and a higher AUC means the scores of attacked and benign windows are better separated.

5.2 Evaluation Results

5.2.1 Validation in Simulator

LCM calibrating its threshold on benign data. The results are as follows. On the 900 missions the detector fires on 887 and misses none: 300 of 300 in S1, 295 of 300 in S2 and 292 of 300 in S3. The remaining 13, five in S2 and eight in S3, activate before the injected onset, from 1.3 s to 186.0 s ahead of it, and are excluded from the detection success evaluation because no attack is present when the alarm fires; they amount to a trace-level spurious activation rate of 1.44%. On the held-out benign split the union stays silent: no alarm on any of the 34 traces, and none in 119,145 member-window decisions, that is 39,715 windows judged by each of the three members. Detection is fast. Over the 887 detections the mean activation delay is 0.498 s, the median 0.500 s, the 90th percentile 0.600 s and the maximum 1.000 s, so every detection lands within one second of onset. By scenario the means are 0.474 s, 0.474 s and 0.547 s with medians 0.425 s, 0.500 s and 0.550 s.

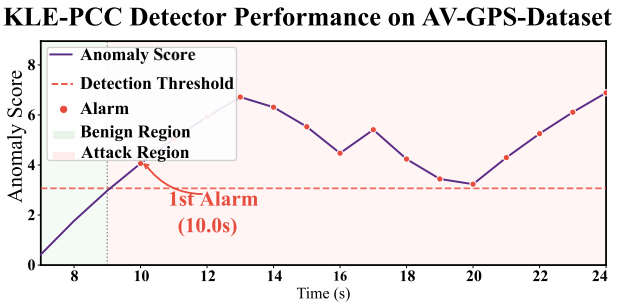


Fig. 2 LCM successfully captures the physical anomalies, triggering the first alarm 1.0 s after the attack onset at $t = 9.0$ s.

Table 2 Detection comparison between LCM and SAVIOR on the 900 attacked missions and 934 attack-free segments, under an identical (self-calibrated) protocol. Trace-level AUC is reported for all detectors; detection rate, false-positive rate (FPR), and mean delay are reported for the operating points. A dash denotes a single-channel variant evaluated at the AUC level only.

Detector	AUC	Detection	FPR	Delay
LCM	**0.9486**	**98.6%**	**1.39%**	**0.498 s**
SAVIOR S_x	0.7537	–	–	–
SAVIOR S_y	0.7479	–	–	–
SAVIOR (union)	0.9924	49.3%	50.64%	2.772 s

We then report discriminability as the trace-level area under the ROC curve (AUC), with the 900 attacked missions as positives and 933 attack-free segments as negatives. Under this protocol LCM reaches an AUC of 0.949 (95% CI $[0.9385, 0.9573]$), confirming that attacked and benign traces are well separated.

5.2.2 Validation on Real-World Data

Simulation alone leaves the question of whether the method survives contact with physical hardware. We therefore also evaluate LCM on the public AV-GPS-Dataset [30], recorded from a rover under real GPS spoofing attacks. Figure 2 shows a representative case, where the alarm comes 1 s after the attack begins. Across the three physical recordings, the detector reaches $\overline{\mathrm{ADD}} = 4\,\mathrm{s}$.

5.3 Comparison Results

We compare LCM against SAVIOR [16], the seminal physics-based detector, under an identical calibration protocol. As shown in Table 2, on single-channel detection, LCM clearly outperforms SAVIOR: it reaches a trace-level AUC of 0.9486, against 0.7537 for SAVIOR's S_x channel and 0.7479

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for its S_y channel across the same 900 missions, .

SAVIOR's fused variant attains a higher raw AUC (0.9924), but at a cost that rules it out in practice: over the 934 attack-free segments it raises false alarms on more than half of them (a 50.6% false-positive rate) while detecting only 49.3% of attacks, with a mean delay of 2.772 s. LCM, in contrast, combines its 0.9486 AUC with a 98.6% detection rate, a 1.39% false-positive rate, and a 0.498 s mean delay, remaining usable on all four axes at once.

6. Stealthy Attacks against LCM

Here we perform stealthy attacks against LCM and ask how much damage an adversary can do while staying below its alarm. A detector is only as trustworthy as its behavior against an attacker who knows how it works, so we take the attacker's side and look for spoofing signals that slip past LCM in a closed loop.

6.1 Threat Model

We assume a white-box attacker who knows the full detection pipeline: the KLE projection, the PCC monitoring with EWMA smoothing, the trained basis vectors, the baseline statistics $(\mu, \sigma_{\text{ewma}})$, and the threshold κ. The attacker adds a perturbation to the sensor measurements to maximize the cumulative positional deviation $\|\delta_t\|$ while keeping the anomaly score below the threshold, $q_k < \kappa$, in every window k. We therefore build slow ramp stealthy attack against this detector, corrupting a single channel to slip past the EWMA smoother.

6.2 Slow Ramp Stealthy Attack

The slow ramp stealthy attack exploits the memory of the EWMA smoother. With a small λ, the update $S_k = \lambda z_k + (1-\lambda)S_{k-1}$ leans heavily on the past, so a perturbation that grows slowly enough barely moves S_k from one step to the next, and the score q_k never accumulates past the threshold. Starting at t_a, the attacker adds a linearly growing offset to one channel,

$$\delta_t^{(\text{channel})} = \min\big(\alpha(t - t_a),\ \delta_{\max}\big), \tag{10}$$

with ramp rate $\alpha > 0$ and saturation $\delta_{\max}$. The spoofed channel $\tilde{y}_t = y_t + \delta_t^{(y)}$ feeds the PID controller, which reads it as a cross-track error and steers to correct it, nudging the vehicle off its lane while the detection score climbs only slowly.

A single mission fixes one spawn point, one speed, and one onset, so its deviation is one sample, not a bound. We therefore repeat the attack over 100 independent missions, drawing the spawn point, target speed $\mathcal{U}(6, 10)$ m/s, and onset $\mathcal{U}(15, 45)$ s per mission (120 s horizon, $\alpha = 0.005$ m/step, $\delta_{\max} = 10$ m).

The detector alarms on 94 of the 100 missions and is evaded on 4 (95% Clopper–Pearson CI [1.1, 9.9]), with a median delay of 41.17 s. Notably, on 93 of the alarms the first channel to fire is `lon_v` (longitude against speed), not the directly attacked `lat_lon`: as the vehicle steers toward the spoofed position, the break in its speed–position relationship is a stronger and earlier signature than the injected offset itself. A monitor watching only the attacked channel would react later, and would underestimate its own exposure. For safety, the quantity that matters is the deviation already accrued when the alarm arrives, since no controller can recover distance lost before it acts. This has a median of 3.759 m and exceeds the 5 m success tolerance on 15 of the 94 alarming missions. The four evaded missions are the most severe, not the mildest (median 7.449 m), so evasion and consequence go together. The binding constraint here is detection delay, not the controller.

Slowing the ramp does not help the adversary. At smaller rates ($\alpha \in \{0.0005, 0.001, 0.002\}$ m/step), the peak deviation falls to a median of 1.73 m at the lowest setting: a ramp slow enough to evade is a ramp that goes nowhere within a mission. That trade-off is what bounds this attack.

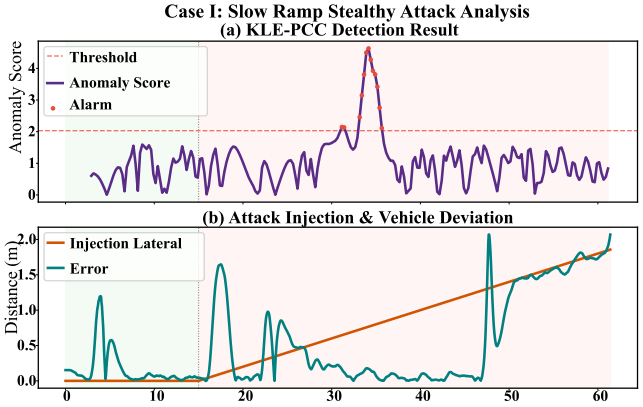


Fig. 3 Closed-loop Slow Ramp stealthy attack analysis. (a) KLE–PCC anomaly score q_k with detection threshold $\kappa = 2.1302$; the first alarm is raised at $t = 31.15$ s. (b) Injected offset magnitude $\|\boldsymbol{\delta}_t\|$ and resulting lateral route deviation e_{lat}; the vehicle deviation exceeds the injection due to closed-loop amplification.

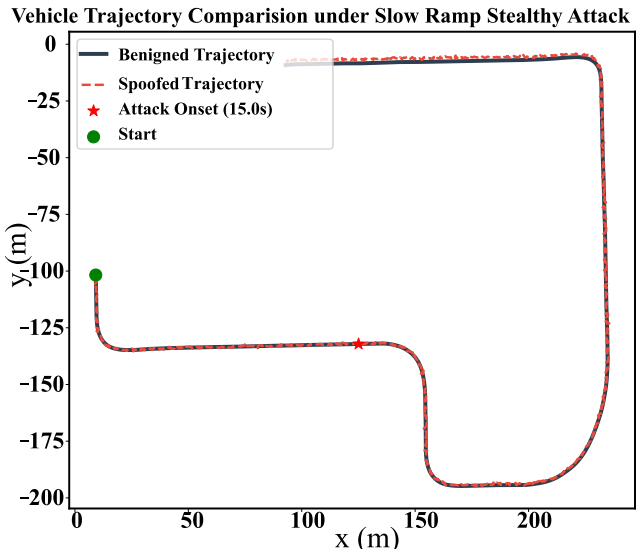


Fig. 4 Vehicle trajectory under the Slow Ramp attack. The true trajectory (solid) and the spoofed trajectory perceived by the controller (dashed) gradually diverge after the attack onset at $t = 15$ s, ultimately causing the vehicle to depart the road.

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Here is one example of the runs. We run this attack on the lateral position channel with onset $t_a = 15\,\mathrm{s}$. In Figure 3(a), the score sits well below the threshold $\kappa = 2.1302$ before the attack, and stays suppressed for a long stretch after it, so the slow ramp does delay detection. But the growing offset keeps pulling the latitude–orientation correlation apart, and the first alarm fires at $t = 31.15\,\mathrm{s}$, a delay of $16.15\,\mathrm{s}$. Figure 3(b) shows the sting: the injected offset reaches $1.856\,\mathrm{m}$, yet the actual lateral deviation reaches $2.069\,\mathrm{m}$. The controller steers toward the spoofed position, so the feedback loop amplifies the attack rather than resisting it. Figure 4 makes this visible: the true and spoofed trajectories are nearly identical through the first half of the mission, then split apart, and the vehicle runs off the road.

7. Discussion

7.1 Limitations

LCM detects a broad range of sensor attacks well, but it has limits worth stating plainly.

Scope limited to inertial and positioning sensors. LCM watches the localization layer: the IMU, GNSS, and wheel encoder. Extending it to perception sensors such as cameras and LiDAR is a different problem, with far higher data volume and semantics that correlation monitoring does not capture.

Dependence on sensor redundancy. LCM detects an attack by catching the inconsistency it creates between sensors, so it needs enough sensors for that inconsistency to show. On a platform with minimal sensing, a corrupted channel may have no peer to disagree with, and the attack can pass unseen. Characterizing how much redundancy is required, and how the detector should behave as that margin thins, is left to future work.

7.2 Future Directions

Several directions follow naturally from this work.

Characterizing the redundancy it needs. LCM detects an attack through the inconsistency it creates between sensors, so it depends on having enough sensors for that inconsistency to appear. A formal account of how much redundancy is required, and how the detector should degrade when sensing is sparse, would tell practitioners when the method applies.

From detection to response. LCM raises an alarm; it does not yet decide what to do next. Pairing it with a state estimator or a recovery controller that acts on its alarms would close the loop from detection to safe control.

Stronger adaptive adversaries. We tested one type of stealthy attack that assume white-box knowledge of the detector. Attacks that adapt online, or coordinate across more than two channels, would probe the limits of correlation monitoring further.

8. Conclusion

We presented LCM, a detector that flags sensor attacks on autonomous vehicles by watching the physical coupling between sensors rather than predicting how the vehicle should behave. Projecting each sensor stream onto its dominant Karhunen–Loève mode and tracking the Pearson correlation of the leading coefficients with an EWMA control chart, LCM needs only benign data to calibrate and assumes nothing about the attack itself.

Across 900 CARLA missions spanning bias, replay, delay, and DoS attacks, LCM detects 887 of the 900 attacked missions with a mean delay of 0.498,s, raises no false alarms on benign data, and achieves an AUC of 0.949. On a real rover under GPS spoofing it reached a mean detection delay of 4 s. We also took the attacker's side: even against an adversary who knows the detector inside out and shapes its injection to stay below the alarm, LCM still caught 94 of 100 attack missions, and the few that slipped through were the most damaging, at a median deviation of 7.449 m.

LCM watches only the inertial and positioning sensors, and it leans on having enough of them for an attack to reveal itself as an inconsistency. How much redundancy that takes, and how the detector should behave when it runs thin, are the questions we take up next.

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References

[1] Becker, J. C. and Simon, A.: Sensor and navigation data fusion for an autonomous vehicle, *Proceedings of the IEEE Intelligent Vehicles Symposium 2000 (Cat. No. 00TH8511)*, IEEE, pp. 156–161 (2000).

[2] Figueiredo, L., Jesus, I., Machado, J. T., Ferreira, J. R. and De Carvalho, J. M.: Towards the development of intelligent transportation systems, *ITSC 2001. 2001 IEEE intelligent transportation systems. Proceedings (Cat. No. 01TH8585)*, IEEE, pp. 1206–1211 (2001).

[3] Pham, M. and Xiong, K.: A survey on security attacks and defense techniques for connected and autonomous vehicles, *Computers Security*, Vol. 109, p. 102269 (online), DOI: https://doi.org/10.1016/j.cose.2021.102269 (2021).

[4] Bertoncello, M. and Wee, D.: Ten ways autonomous driving could redefine the automotive world, *McKinsey & Company*, Vol. 6 (2015).

[5] Jahan, F., Sun, W., Niyaz, Q. and Alam, M.: Security modeling of autonomous systems: A survey, *ACM Computing Surveys (CSUR)*, Vol. 52, No. 5, pp. 1–34 (2019).

[6] of California Department of Motor Vehicles, S.: Autonomous Vehicle Collision Reports (n.d.). Accessed: January 18, 2025 from `https://www.dmv.ca.gov/portal/vehicle-industry-services/autonomous-vehicles/autonomous-vehicle-collision-reports/`.

[7] Lee, Gober Reyna, P. I. A.: Examining Autonomous Car Accidents and Statistics (n.d.). Accessed: January 19, 2025 from `https://www.lgrlawfirm.com/blog/examining-autonomous-car-accidents-and-statistics-2/`.

[8] Gao, C., Wang, G., Shi, W., Wang, Z. and Chen, Y.: Autonomous Driving Security: State of the Art and Challenges, *IEEE Internet of Things Journal*, Vol. 9, No. 10, pp. 7572–7595 (online), DOI: 10.1109/JIOT.2021.3130054 (2022).

[9] Talaei Khoei, T., Al Shamaileh, K., Devabhaktuni, V. K. and Kaabouch, N.: Performance analysis of capsule networks for detecting GPS spoofing attacks on unmanned aerial vehicles, *International Journal of Information Security*, Vol. 24, No. 1, p. 62 (2025).

[10] Sun, J., Cao, Y., Chen, Q. A. and Mao, Z. M.: Towards robust {LiDAR-based} perception in autonomous driving: General black-box adversarial sensor attack and countermeasures, *29th USENIX Security Symposium (USENIX Security 20)*, pp. 877–894 (2020).

[11] Jeong, J., Kim, D., Jang, J.-H., Noh, J., Song, C. and Kim,

© 2026 Information Processing Society of Japan

Y.: Un-Rocking Drones: Foundations of Acoustic Injection Attacks and Recovery Thereof., *NDSS* (2023).

[12] Gao, M., Zhang, L., Shen, L., Zou, X., Han, J., Lin, F. and Ren, K.: KITE: Exploring the practical threat from acoustic transduction attacks on inertial sensors, *Proceedings of the 20th ACM Conference on Embedded Networked Sensor Systems*, pp. 696–709 (2022).

[13] Gao, M., Zhang, L., Shen, L., Zou, X., Han, J., Lin, F. and Ren, K.: Exploring practical acoustic transduction attacks on inertial sensors in MDOF systems, *IEEE Transactions on Mobile Computing*, Vol. 23, No. 5, pp. 3539–3557 (2023).

[14] Zhang, J., Wang, Y., Tu, Y., Rampazzi, S., Lin, Z., Lee, I. and Hei, X.: ADC-Bank: Detecting Acoustic Out-of-Band Signal Injection on Inertial Sensors, *International Conference on Security and Privacy in Cyber-Physical Systems and Smart Vehicles*, Springer, pp. 53–72 (2023).

[15] Choi, H., Lee, W.-C., Aafer, Y., Fei, F., Tu, Z., Zhang, X., Xu, D. and Deng, X.: Detecting attacks against robotic vehicles: A control invariant approach, *Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security*, pp. 801–816 (2018).

[16] Quinonez, R., Giraldo, J., Salazar, L., Bauman, E., Cardenas, A. and Lin, Z.: {SAVIOR}: Securing autonomous vehicles with robust physical invariants, *29th USENIX security symposium (USENIX Security 20)*, pp. 895–912 (2020).

[17] Abdo, A., Malek, S. M. B., Zhao, X. and Abu-Ghazaleh, N.: AVMON: Securing Autonomous Vehicles by Learning Control Invariants and Residual Prediction, *VehicleSec* (2024).

[18] Dash, P., Li, G., Chen, Z., Karimibiuki, M. and Pattabiraman, K.: Pid-piper: Recovering robotic vehicles from physical attacks, *2021 51st Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, IEEE, pp. 26–38 (2021).

[19] Fu, K. and Xu, W.: Risks of trusting the physics of sensors, *Communications of the ACM*, Vol. 61, No. 2, pp. 20–23 (2018).

[20] Watanabe, S.: Karhunen-Loève Expansion and Factor Analysis, *Proc. 4th Prague Conf. on Information Theory, Statistical Decision Functions, Random Processes*, Prague, Czechoslovakia, Publishing House of the Czechoslovak Academy of Sciences (1967).

[21] Watanabe, S.: Evaluation and Selection of Variables in Pattern Recognition, *Computer and Information Sciences II* (1967).

[22] Chien, Y.-T. and Fu, K.-S.: On the Generalized Karhunen-Loève Expansion, *IEEE Transactions on Information Theory*, Vol. 13, No. 3, pp. 518–520 (online), DOI: 10.1109/tit.1967.1054021 (1967).

[23] Sedgwick, P.: Pearson's Correlation Coefficient, *BMJ*, Vol. 345, p. e4483 (online), DOI: 10.1136/bmj.e4483 (2012).

[24] Benesty, J., Chen, J., Huang, Y. and Cohen, I.: Pearson Correlation Coefficient, *Noise Reduction in Speech Processing*, Springer, Berlin, Germany, pp. 1–4 (2009).

[25] Barnard, G. A.: Control Charts and Stochastic Processes, *Journal of the Royal Statistical Society: Series B (Methodological)*, Vol. 21, No. 2, pp. 239–257 (online), DOI: 10.1111/j.2517-6161.1959.tb00336.x (1959).

[26] Hunter, J. S.: The Exponentially Weighted Moving Average, *Journal of Quality Technology*, Vol. 18, No. 4, pp. 203–210 (online), DOI: 10.1080/00224065.1986.11979014 (1986).

[27] Lucas, J. M. and Saccucci, M. S.: Exponentially Weighted Moving Average Control Schemes: Properties and Enhancements, *Technometrics*, Vol. 32, No. 1, pp. 1–12 (online), DOI: 10.1080/00401706.1990.10484583 (1990).

[28] Roberts, S. W.: Control Chart Tests Based on Geometric Moving Averages, *Technometrics*, Vol. 42, No. 1, pp. 97–101 (online), DOI: 10.1080/00401706.2000.10485986 (2000).

[29] Dosovitskiy, A., Ros, G., Codevilla, F., López, A. M. and Koltun, V.: CARLA: An Open Urban Driving Simulator, *Proc. Conf. on Robot Learning (CoRL)*, PMLR, pp. 1–16 (online), available from ⟨http://proceedings.mlr.press/v78/dosovitskiy17a.html⟩ (2017).

[30] Abrar, M. M., Youssef, A., Islam, R., Satam, S., Latibari, B. S., Hariri, S., Shao, S., Salehi, S. and Satam, P.: GPS-IDS: An Anomaly-based GPS Spoofing Attack Detection Framework for Autonomous Vehicles, *arXiv preprint arXiv:2405.08359*, (online), DOI: 10.48550/arXiv.2405.08359 (2024).

[31] Chen, X., Lei, M., Saunier, N. and Sun, L.: Low-Rank Autoregressive Tensor Completion for Spatiotemporal Traffic Data Imputation, *IEEE Transactions on Intelligent Transportation Systems*, Vol. 23, No. 8, pp. 12301–12310 (online), DOI: 10.1109/tits.2021.3113608 (2022).

[32] Zhou, Z., Wu, Z., Bolli, F., Boutteau, R., Yang, F., Timofte, R., Ginhac, D. and Delbruck, T.: Steering Prediction via a Multi-Sensor System for Autonomous Racing, *Proc. IEEE Int. Conf. on Robotics and Automation (ICRA)*, pp. 8612–8618 (online), DOI: 10.1109/icra55743.2025.11128135 (2025).

[33] Ephraim, Y. and Van Trees, H. L.: A Signal Subspace Approach for Speech Enhancement, *IEEE Transactions on Speech and Audio Processing*, Vol. 3, No. 4, pp. 251–266 (online), DOI: 10.1109/89.397090 (1995).

[34] Hermus, K., Wambacq, P. and Van Hamme, H.: A Review of Signal Subspace Speech Enhancement and Its Application to Noise Robust Speech Recognition, *EURASIP Journal on Advances in Signal Processing*, Vol. 2007, No. 1, p. 45821 (online), DOI: 10.1155/2007/45821 (2007).

[35] Benesty, J., Chen, J. and Huang, Y.: On Noise Reduction in the Karhunen-Loève Expansion Domain, *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, pp. 25–28 (online), DOI: 10.1109/icassp.2009.4959511 (2009).

[36] Sharma, P., Petit, J. and Liu, H.: Pearson Correlation Analysis to Detect Misbehavior in VANET, *Proc. 2018 IEEE 88th Vehicular Technology Conference (VTC-Fall)*, IEEE, pp. 1–5 (online), DOI: 10.1109/vtcfall.2018.8690964 (2018).

[37] Guo, F., Wang, Z., Du, S., Li, H., Zhu, H., Pei, Q., Cao, Z. and Zhao, J.: Detecting Vehicle Anomaly in the Edge via Sensor Consistency and Frequency Characteristic, *IEEE Transactions on Vehicular Technology*, Vol. 68, No. 6, pp. 5618–5628 (online), DOI: 10.1109/tvt.2019.2907692 (2019).

[38] Won, J., Seo, S.-H. and Bertino, E.: A Secure Shuffling Mechanism for White-Box Attack-Resistant Unmanned Vehicles, *IEEE Transactions on Mobile Computing*, Vol. 19, No. 5, pp. 1023–1039 (online), DOI: 10.1109/tmc.2019.2903048 (2020).

[39] Xu, C., Zheng, F. and Guo, G.: Stealthy False Data Injection Attacks in Multi-Channel Vehicular Communication: White-Box and Gray-Box Strategies, *IEEE Transactions on Vehicular Technology*, Vol. 74, No. 9, pp. 13402–13416 (online), DOI: 10.1109/tvt.2025.3560185 (2025).

[40] Hua, J., Liu, R. and Hao, F.: Two-Channel False Data Injection Attacks on Multi-Sensor Remote State Estimation, *Asian Journal of Control*, Vol. 25, No. 5, pp. 3776–3791 (online), DOI: 10.1002/asjc.3067 (2023).

[41] Mei, S. and Zhu, X.: Using Machine Teaching to Identify Optimal Training-Set Attacks on Machine Learners, *Proc. AAAI Conf. Artificial Intelligence*, Vol. 29, No. 1, (online), DOI: 10.1609/aaai.v29i1.9569 (2015).

[42] Shafahi, A., Huang, W. R., Najibi, M., Suciu, O., Studer, C., Dumitras, T. and Goldstein, T.: Poison Frogs! Targeted Clean-Label Poisoning Attacks on Neural Networks, *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 31, pp. 6106–6116 (2018).

[43] Lu, P., Zhang, L., Liu, M., Sridhar, K., Sokolsky, O., Kong, F. and Lee, I.: Recovery from Adversarial Attacks in Cyber-Physical Systems: Shallow, Deep, and Exploratory Works, *ACM Computing Surveys*, Vol. 56, No. 8, pp. 1–31 (online), DOI: 10.1145/3653974 (2024).

[44] Pike, L., Hickey, P., Elliott, T., Mertens, E. and Tomb, A.: TrackOS: A Security-Aware Real-Time Operating System, *Proc. Runtime Verification: 16th Int. Conf. (RV 2016)*, Madrid, Spain, Springer, pp. 302–317 (2016).

[45] Clements, A. A., Almakhdhub, N. S., Saab, K. S., Srivastava, P., Koo, J., Bagchi, S. and Payer, M.: Protecting Bare-Metal Embedded Systems with Privilege Overlays, *Proc. IEEE Symposium on Security and Privacy (SP)*, pp. 289–303 (online), DOI: 10.1109/sp.2017.37 (2017).

[46] Cho, M., Cao, Y., Zhou, Z. and Mao, Z. M.: ADoPT: LiDAR Spoofing Attack Detection Based on Point-Level Temporal Consistency, *Proc. British Machine Vision Conference (BMVC)*, pp. 486–496 (online), available from ⟨http://proceedings.bmvc2023.org/486/⟩ (2023).

[47] Hau, Z., Demetriou, S. and Lupu, E. C.: Using 3D Shadows to Detect Object Hiding Attacks on Autonomous Vehicle Perception, *Proc. IEEE Security and Privacy Workshops (SPW)*, Los Alamitos, CA, USA, IEEE Computer Society, pp. 229–235 (online), DOI: 10.1109/spw54247.2022.9833890 (2022).

[48] Zhu, Y., Miao, C., Xue, H., Li, Z., Yu, Y., Xu, W., Su, L. and Qiao, C.: TileMask: A Passive-Reflection-Based Attack Against mmWave Radar Object Detection in Autonomous

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Driving, *Proc. ACM SIGSAC Conf. on Computer and Communications Security (CCS)*, New York, NY, USA, Association for Computing Machinery, pp. 1317–1331 (online), DOI: 10.1145/3576915.3616661 (2023).

[49] Chen, P., Zhao, X., Zeng, L., Liu, L., Liu, S., Sun, L., Li, Z., Chen, H., Liu, G., Qiao, Z., Qu, Y., Xu, D., Li, L. and Li, L.: A Review of Research on SLAM Technology Based on the Fusion of LiDAR and Vision, *Sensors*, Vol. 25, No. 5, p. 1447 (online), DOI: 10.3390/s25051447 (2025).

[50] Wu, Y., Kuang, J., Niu, X., Stachniss, C., Klingbeil, L. and Kuhlmann, H.: Wheel-GINS: A GNSS/INS integrated navigation system with a wheel-mounted IMU, *IEEE Transactions on Intelligent Transportation Systems*, Vol. 26, No. 5, pp. 6891–6903 (online), DOI: 10.1109/tits.2025.3527815 (2025).

[51] Zhang, L., Lu, P., Kong, F., Chen, X., Sokolsky, O. and Lee, I.: Real-Time Attack-Recovery for Cyber-Physical Systems Using Linear-Quadratic Regulator, *ACM Transactions on Embedded Computing Systems*, Vol. 20, No. 5s, pp. 1–24 (online), DOI: 10.1145/3477010 (2021).

[52] Nauta, T., Sandberg, H. and Maggio, M.: Stealthy Computational Delay Attacks on Control Systems, *Proceedings of the ACM/IEEE 16th International Conference on Cyber-Physical Systems (with CPS-IoT Week 2025)*, ICCPS '25, New York, NY, USA, Association for Computing Machinery, (online), DOI: 10.1145/3716550.3722013 (2025).

[53] Dibaji, S. M., Pirani, M., Flamholz, D. B., Annaswamy, A. M., Johansson, K. H. and Chakrabortty, A.: A systems and control perspective of CPS security, *Annual reviews in control*, Vol. 47, pp. 394–411 (online), DOI: 10.1016/j.arcontrol.2019.04.011 (2019).

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